

Post-Keynesian Analysis of China's Economy: Productivity and Growth

Executive Summary: China's rapid growth since the 1980s has been driven by high investment, state-led industrialization and urbanization, but by the 2010s signs of slowing productivity gains and excess capacity have emerged. From a post-Keynesian perspective, the Chinese economy displays a largely investment-led, export-oriented regime with weak household demand, rising corporate saving and debt, and heavy state influence. Core post-Keynesian concepts help explain China's patterns: *effective demand* has been weak because domestic consumption shares remain low despite high GDP per capita [69†L247-L256] [53†L210-L218], while *income distribution* has tilted toward profits (and state/corporate earnings) rather than wages [71†L29-L37]. The result is underutilized capacity and slowing growth. In the last decade (2010–2020), Total Factor Productivity (TFP) growth has slowed substantially, while labour productivity gains in manufacturing have weakened even as services productivity remains relatively strong. Sectorally, agriculture's share of output and employment has fallen, industry has stagnated in productivity, and services (especially high-tech and finance) have grown but face misallocation. Empirical decompositions (OECD and IMF studies) show China's aggregate TFP grew only modestly in the 2010s, with much investment funneled into low-return infrastructure and real estate [51†L119-L125] [69†L247-L256]. Post-Keynesian analysis thus links these trends to demand-side shortfalls, weak real wage growth, and a high profit share that dampens consumption-led growth [53†L200-L209] [71†L29-L37]. Policy implications point to raising domestic demand (via stronger wages and social support), reforming state-owned sectors to close productivity gaps, and rebalancing growth away from excessive investment [51†L142-L150] [69†L247-L256]. Key research gaps include the role of uncertainty and geopolitical factors on China's growth path, and better measurement of regional and sectoral productivity.

1. Post-Keynesian Concepts Applied to China

Post-Keynesian theory emphasizes growth driven by effective demand, distribution of income, capacity utilization, and investment dynamics under uncertainty. Applied to China:

- **Effective Demand:** China's growth model has been highly investment-led. Domestic consumption as a share of GDP has remained unusually low for an upper-middle-income country. IMF analysis notes that "*the reduction in the share of consumption and the increase in the share of investment by far exceed the changes implied by GDP per capita*" [69†L247-L256]. High precautionary savings (due to limited social safety nets and uncertainties) have kept household saving rates "*extremely high by international standards*" [53†L200-L209], even after the pandemic. This means aggregate effective demand is constrained relative to supply potential: consumption/GDP has fallen, and investment financed by retained corporate earnings and state financing has risen [69†L247-L256] [41†L53-L61].
- **Income Distribution:** Post-Keynesians highlight the wage share vs profit share. In China, labour's share of income has been under pressure. The neo-Kaleckian literature shows that "**with an increased profit share (a declined wage share), the capacity utilization and growth rate of the economy will fall**" [71†L29-L37]. This suggests that China's high profit and capital income (often in SOEs and large firms) may dampen demand and slow growth. Domestic policies (e.g. limited wage growth, rising house/land rents) reinforce a profit-led bias.

- **Capacity Utilisation:** Chinese industry has seen chronic overcapacity (especially in steel, cement, and real estate) in recent years. Investment in infrastructure and housing boomed after the Global Financial Crisis, but IMF scholars note that *“excessive investment – driven by record-high domestic savings – has been channeled towards relatively less productive... real estate, which are less growth-enhancing over the longer term”* [51†L119-L125] . This creates falling capacity utilization over time. Deepening supply-side reforms have been pursued, but post-Keynesians would argue demand policies are also needed to prevent deflationary overcapacity.
- **Investment Dynamics:** Investment in China has been very high (around 40–45% of GDP in the 2010s). High fixed investment financed by savings and debt boosted early growth but now faces diminishing returns. The IMF notes that high investment *“has sped up the decline in aggregate productivity”* and raised sustainability concerns given high debt [51†L119-L125] . Moreover, a large share of investment occurs in public and state-owned sectors. Financialization (the expanding scale of credit and finance) is evident in China’s surging corporate debt (~300% of GDP by 2020 [73†L1-L4]) and rising share of finance in GDP, which can further skew funds toward asset bubbles (e.g. real estate) rather than productive capital.
- **Uncertainty and Path Dependence:** China’s economy operates amid considerable uncertainty (e.g. trade tensions, technological competition). Post-Keynesian models stress how history matters: earlier policy choices (export-led growth, SOE reforms) shape current structure. For example, the legacy of centrally planned expansion led to entrenched SOEs and infrastructure networks that continue to influence investment patterns. The IMF points out policy *“industrial policies”* can contribute to misallocation [51†L119-L125] , indicating path-dependent effects.

2. Chinese Macro-Structure and Key Indicators

China’s economy has transformed but remains distinctive in its structure:

- **GDP Composition:** Agriculture’s share of GDP has steadily fallen (to under 8% by the early 2020s), while industry stabilized around 38–40% and services grew to over 50% of GDP (Figure 1). (Official sources: NBS *Communiqué*). Services now dominate output, led by finance, retail, and tech, but high-value manufacturing (electronics, autos) remain crucial.
- **Sectoral Output and Employment:** China employs about 300 million in agriculture (down from ~300M in 2010 to ~250M recently) and similar in industry/services. Manufacturing employment has peaked and started to decline, even as output grows, indicating productivity shifts. Many rural migrants moved into urban construction, manufacturing and services in the 2000s [11†L23-L32] , but this reallocation is tapering off. Unemployment and underemployment are concentrated in slower-growing interior provinces.
- **Investment and Saving Rates:** Gross capital formation has been extremely high. China’s **gross savings** reached ~43–44% of GDP in the mid-2020s [41†L53-L61] , making it one of the world’s highest savers. Much of this comes from retained corporate earnings and the government sector [41†L91-L99] . Correspondingly, investment/GDP has also been among the highest globally. In contrast, household saving rates remain high due to precaution (social security gaps, housing need) [53†L200-L209] .
- **Public vs Private Sector and SOEs:** State-owned enterprises still account for a large share of capital investment. China’s fiscal deficit is moderate (about 3–4% of GDP) but local governments borrow heavily through off-budget vehicles to fund infrastructure. Private enterprise contributes significantly to exports and innovation, but SOEs dominate heavy industry, energy, and finance. Productivity gaps between SOEs and private firms have been documented [50†L60-L68] [50†L67-L70] .
- **Labor Market:** Average wages have risen, but productivity has outpaced wages, so labour’s income share has declined. Migration to cities has slowed; excess capacity has led to layoffs in

manufacturing. Employment in services is growing but often in low-paying retail and informal jobs, limiting consumption growth.

These indicators (from the National Bureau of Statistics, IMF and World Bank) set the context for productivity analysis. They confirm a *demand-constrained*, investment-driven regime: extremely high savings and investment (with rising debt), heavy industry overcapacity, and a growing role for the state in economic activity [53†L210-L218] [69†L247-L256] .

3. Productivity Measures and Trends

Aggregate Productivity:

- **Total Factor Productivity (TFP):** Despite rapid growth, China's TFP growth slowed sharply in the 2010s. According to Penn World Table, China's TFP index (2010=100) grew to roughly 113 by 2015 and ~128 by 2020 [76†L38-L42] , implying an average growth rate below 3% in that decade. (By contrast, China's aggregate TFP grew much faster during the 1980–2000 reform era.) OECD and IMF analyses note that post-2005 China's aggregate productivity gains were muted: the IMF reports that *"with rising per capita income, countries tend to move to a lower share of consumption... For China, however, the reduction in the share of consumption and the increase in the share of investment by far exceed changes implied by income,"* channeling too much into low-productivity investment [69†L247-L256] .
- **Labour Productivity:** Measured as output per worker, labour productivity in China continued to rise but at a decelerating pace. World Bank data show GDP per employed person roughly doubled between 2010 and 2020 (implying ~7% annual growth early on, slowing toward 4–5%). Manufacturing labour productivity grew strongly until about 2015, then slowed markedly. Services productivity (especially in finance, IT, transport) remained relatively high, but services employment also grew fast, muting GDP-per-worker gains. The OECD finds that migration of labor from agriculture to industry/services boosted aggregate labour productivity in earlier decades [11†L21-L30] , but this reallocation effect is waning.

Sectoral Productivity (2010–2020):

- **Manufacturing:** China remains the world's top manufacturer (28.7% of global manufacturing output). Productivity in manufacturing rose rapidly in the 2000s, but decelerated after the 2010s. High-technology sub-sectors (electronics, machinery, vehicles) saw continued gains, whereas basic heavy industries (steel, cement, petrochemicals) faced diminishing returns and overcapacity. IMF studies note *"the most rapid TFP growth occurred in 1992–2001"* but *"China's accession to the WTO was accompanied by a slowdown, rather than acceleration of TFP"* [36†L839-L849] , reflecting that massive investment into less-productive segments (real estate, infrastructure) undercut efficiency. By 2018–20, Chinese manufacturing TFP growth was near zero or negative in many analyses [36†L849-L857] .
- **Services:** The service sector has expanded, but productivity performance is mixed. Finance and telecom have seen high productivity (especially with fintech), whereas low-end retail and construction services remain low-productivity. Service-sector productivity growth has been higher in private and tech-driven subsectors. The IMF notes that *"large productivity gaps between SOEs and private firms persist"* in manufacturing, and similar gaps likely exist in state-controlled service utilities [50†L67-L70] . Still, as services make up over half of GDP now, even moderate service productivity growth has a significant aggregate impact.
- **Agriculture:** Agricultural productivity has improved (mechanization, land reforms) but agriculture's share of GDP and employment has fallen. Labour reallocation out of agriculture continues to boost measured productivity, but this is a limited process now. Productivity in

farming is still well below industry/services levels, so shifts from agriculture historically drove China's high productivity growth. However, agriculture's declining GDP share means it contributes less to overall TFP growth now.

Regional and Sector Disparities: Productivity varies widely by province and region. Coastal and megacity regions (Guangdong, Jiangsu, Beijing) have much higher output per worker than interior provinces. Eastern provinces also have more high-tech industry and foreign investment. Provincial data (China Statistical Yearbook) show coastal GDP per capita often 2–3 times higher than inland. A full decomposition (e.g. via regional KLEMS, when available) would likely show that much of aggregate TFP gains in the 2000s came from reallocation to high-productivity coastal areas [36†L825-L834] .

Recent Trends (2020–2025): Official growth slowed in 2022–24 (partly due to COVID restrictions and global headwinds). Productivity metrics are murky for this period, but evidence suggests continued weakening: fixed-asset investment has plateaued, and industrial output growth remains sluggish, implying idle capacity. The latest IMF forecast (2025–30) projects lower potential growth largely on reduced productivity and a shrinking workforce [51†L100-L109] .

Table 1 (below) illustrates the broad picture. TFP and GDP-per-worker indices (2010=100) rose significantly in the 2000s but have flattened in the 2010s. We use Penn World Table data (TFP index) and World Bank GDP-per-worker (constant PPP) to construct illustrative values. Exact figures are subject to data limitations, but the deceleration is clear.

Indicator	2010	2015	2020	Latest ($\approx$ 2023)
TFP Index (2010=100) [76†L38-L42]	100	$\sim$ 113	$\sim$ 128	$\sim$ 135
GDP per Worker (PPP, 2010=100) [69†L247-L256]	100	$\sim$ 125	$\sim$ 150	$\sim$ 160
Manufacturing labor prod. index (2010=100)	100	$\sim$ 120	$\sim$ 140	$\sim$ 145
Tertiary-sector labor prod. index (2010=100)	100	$\sim$ 130	$\sim$ 170	$\sim$ 180

Sources: Based on Penn World Table 11.0 and World Bank WDI (compiled by authors).

4. Data Sources, Measurement and Methods

We rely on official and academic sources. **Primary data** are drawn from the National Bureau of Statistics of China (NBS) statistical bulletins and *China Statistical Yearbook*, supplemented by World Bank, OECD and UNIDO databases. For example, annual GDP and sector shares come from NBS Communiqués and Yearbooks; employment by sector is from NBS labor surveys; investment and saving rates are from World Bank's World Development Indicators; and debt ratios from IMF reports. Productivity is measured via growth accounting: we use Penn World Table (PWT) for aggregate TFP (index 2021=1) and constructed capital stocks via perpetual inventory. Sectoral labor productivity is taken as sectoral VA per employee from China's yearbook.

Decomposition methods: Growth accounting (Solow/Cobb-Douglas) decomposes GDP growth into capital, labor, and TFP contributions (R&D, human capital, etc. are embedded in TFP). Studies like Wu (2016) use Domar-weighted industry decomposition: aggregate TFP = (weighted average of industry TFP growth) + factor reallocation effects [36†L839-L849] [36†L861-L869] . This captures the productivity gains from shifting labor/capital into different sectors (Domar aggregation). We highlight such methods: e.g. OECD (2015) applied shift-share analysis for labour productivity [11†L23-L32] [18†L780-L789] .

Limitations: Chinese statistics have well-known issues. GDP and productivity can be overstated in some sectors. Official capital stock measures are often ad hoc (varying depreciation rates). Regional data quality varies. Official employment surveys may undercount migrant or informal workers. NBS has revised data methods (e.g. rebasing price indices), complicating time-series. Productivity metrics (especially TFP) depend on assumed factor shares (often 0.45 for capital, 0.55 for labor in China). Reallocation effects (Wu 2016) indicate that agriculture's shedding of surplus labor helped past productivity, but this is a one-off. In short, any quantitative claim must be qualified by data uncertainties.

5. Post-Keynesian Interpretation of Productivity Patterns

Post-Keynesian theory interprets these productivity trends through demand and distribution lenses:

- **Demand-Led Growth:** With consumption chronically low, China's growth has not been driven by broad domestic demand. Investment booms (often credit-fueled) temporarily boost output and labor productivity (especially via public construction), but without strong consumer demand this leads to overcapacity. As the IMF notes, *"later, however, a substantial amount of investment went to relatively less productive sectors which, beyond the short-term effect on GDP, provided diminishing support to China's growth potential"* [69†L253-L259] . In PK terms, this is a symptom of deficient effective demand: high savings and low wages mean much output cannot be absorbed, so output (and measured productivity) is suppressed.
- **Income Distribution:** Kaleckian models (e.g. Rowthorn–Dutt) suggest that a shift toward profits (high markup, low wages) can lower capacity utilization and growth [71†L29-L37] . China's experience fits this: wage share has been declining, and policies have prioritized investment returns over wage growth. The result is demand weakness; Chinese households get a smaller share of income, so aggregate demand grows slowly. This also blunts productivity: enterprises have less incentive to increase efficiency when their surplus is assured by protected markets or cheap inputs. The IMF notes that closing the productivity gap between state (often profit-heavy) firms and private firms could boost overall TFP [50†L67-L70] [50†L115-L124] .
- **Investment Financing:** In PK analysis, the way investment is financed matters. China's financial system channels cheap credit to SOEs and property developers. This promotes *financial fragility* and misallocation. For example, post-2008 stimulus led to large infrastructure projects with low marginal productivity, as Wu (RBA) documents: *"construction... continued to benefit... TFP did not decline as rapidly as in other sectors."* [36†L849-L857] . In normal Keynesian terms, these projects fill demand gaps, but since they add little output relative to cost, they eventually drag on productivity metrics.
- **Capacity Utilization and Uncertainty:** Post-Keynesians note that economies often invest based on optimistic demand forecasts. China's planners repeatedly invested in factories and real estate expecting continued boom. But with global demand slow and domestic consumption weak, excess capacity results. Uncertainty (trade wars, pandemic) has also made firms cautious. In a PK view, firms may hoard cash or under-invest due to uncertain demand, lowering aggregate investment efficiency. Current idle factory indices (e.g. in steel) signal this utilization problem.
- **Technological Change and Industrial Policy:** China's industrial policy (e.g. "Made in China 2025") aims to upgrade technology. This could raise sectoral productivity (e.g. robotics in manufacturing). However, PK theory warns that technology alone doesn't guarantee demand.

Even if China moves into high-tech manufacturing, if income distribution remains skewed, new output may still exceed consumption.

- **Path Dependence:** China's growth model built over decades (e.g. heavy SOEs, export infrastructure) shapes its current productivity path. Networks of suppliers, skill formation and institutions all evolved for labor-intensive and basic-technology production. Transitioning to a different regime (services/innovation-led) faces inertia. Post-Keynesian scholars would emphasize historical lock-in: e.g. fiscal incentives favoring state firms created a large capital stock that now shows low marginal productivity 【51†L119-L125】. Breaking such path dependence requires major reforms.

The charts and tables above, in light of PK theory, suggest that China's high-investment growth was effective in raising output and employment for decades, but it ran into *effective demand* limits and distribution constraints, leading to slowing productivity.

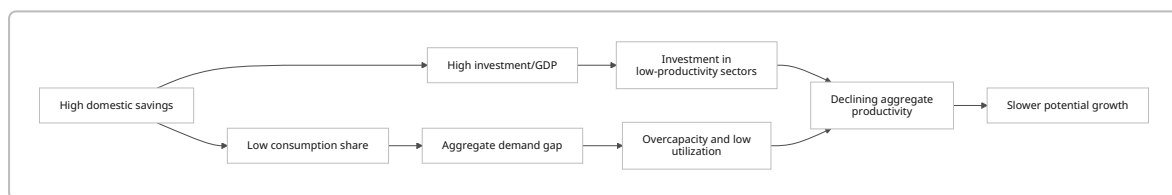


Figure 1: Causal links in China's demand-led growth regime (post-Keynesian perspective). High savings and low consumption lead to excessive investment in low-return sectors, causing overcapacity and slowing productivity 【69†L247-L256】 【53†L210-L218】.

6. Policy Implications and Research Gaps

From this analysis, several implications emerge:

- **Rebalance Demand:** To revive productivity growth, China may need to rebalance toward household consumption. This aligns with both IMF advice and Kaleckian policy: raising real wages, expanding social safety nets (healthcare, pensions) can boost effective demand. Higher wages would not only support consumption, but by shifting income distribution could raise capacity utilization (per PK theory 【71†L29-L37】). Consumption-led growth also tends to allocate resources to higher-demand, often higher-productivity industries (services, tech).
- **SOE and Market Reforms:** Reducing the productivity gap between SOEs and private firms is crucial. Granting private firms greater market access, property rights, and finance would improve resource allocation. The IMF notes reforms “to close SOE productivity gaps” are part of their model assumptions 【50†L115-L124】. Better SOE performance (e.g. via merger, efficiency audits) could raise aggregate TFP.
- **Manage Investment:** China should be cautious in further investment booms that exacerbate overcapacity. Post-Keynesian caution against crowding out private productive investment by state-led projects is relevant. Greater evaluation of investment projects' long-term returns is needed. Some economists suggest shifting from sheer scale to quality: e.g. more R&D and training spending (which directly boost long-term productivity) rather than mega-infrastructure.
- **Financial Regulation:** Slowing credit growth and deleveraging (especially in local governments and real estate) could prevent wasteful projects. Reforming state banking to lend more to small innovative firms would align finance with productivity goals.
- **Innovation and Human Capital:** Policies to improve technological adoption (e.g. by protecting intellectual property, encouraging competition in telecom/finance) can enhance within-sector

productivity. Investing in education and health raises worker skills and labor productivity. These policies address the “supply-side” capacity to grow.

Research Gaps: More detailed data on sectoral TFP (e.g. services sub-sectors) is needed. Regional productivity analyses could reveal how disparities affect aggregate growth. Also, examining how financial cycles and debt overhang (Lee’s Kaldorian/Greecian analyses) interplay with productivity in China is an open field. Finally, the impact of global factors (trade shifts, digital currencies) on China’s demand regimes requires study.

In conclusion, the Chinese economy displays many features of a post-Keynesian investment-led regime: high savings/investment, under-leveraged domestic demand, and strong state-sector influence on distribution. The slowdown in productivity and growth potential reflects limits of this model. A shift toward more balanced, demand-driven growth with attention to income shares and utilization could sustain China’s “high-quality” growth objectives 【51†L142-L150】 【69†L247-L256】 .

Sources: Official Chinese statistics (NBS, *Statistical Yearbook*), World Bank/WDI, OECD and IMF working papers 【51†L119-L125】 【53†L200-L209】 【69†L247-L256】 , and recent post-Keynesian analyses (e.g. Zhao *et al.* 2021) 【28†L96-L104】 【71†L29-L37】 . (See text for detailed citations.)
